FOREIGN TRADE UNIVERSITY

VIETNAM-JAPAN COLLEGE OF BUSINESS AND INNOVATION

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GROUP REPORT

Group 1 - Advanced Business Analytics K62 DB

Asking-Price Estimation and Market Intelligence for Vietnam's Secondary Housing Market

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I. EXECUTIVE SUMMARY

Buyers, brokers and lenders in Hanoi and Ho Chi Minh City (HCMC) have no consistent public reference for judging the asking price of an independently listed resale home. This project builds a platform on 5,268 clean mogi.vn listings linked to 34,321 OpenStreetMap points. It has three engines: a Fair Value Engine (asking-price reference with a 90% range), a Stale Listing Radar (ranks listings likely to stay online more than 180 days) and a Market Map (four district segments).

On unseen homes, the regression reaches R^2 of 0.850, the Radar reaches AUC of 0.81 and the 90% range covers 90.5% of homes. The RMSE and district-separation targets are not met. Only 53.9% of estimates fall within ±20% of the asking price, so the tool supports a broker but cannot replace an appraiser.

The base case nets VND 9.7 bn a year (4.4 times run cost), ranging from −2.3 bn to +40.9 bn, and broker adoption drives the result. We recommend a 12-month rollout with go or no-go gates at months 3, 7 and 12. Results are asking-price references, not valuations for lending, until they are calibrated on transaction prices.

II. SOLUTION DESIGN

1. Problem Statement

Mogi.vn provides area-level asking prices but lacks listing-level valuations, stale-listing alerts and district comparisons (Mogi.vn, 2026; Brands Vietnam, n.d.). Existing tools mainly serve apartment projects or banks (Báo Pháp luật, 2022; e27, 2025). As a result, brokers, lenders and managers lack reliable references for pricing, prioritisation and market decisions.

To address this gap, our platform combines three features: Fair Value Engine for price estimation, Stale Listing Radar for identifying listings needing review, and Market Map for district comparisons. Since estimates rely on asking prices rather than actual transaction prices, human review remains necessary.

2. Data

The prototype uses real public data: sale listings from mogi.vn and map data from OpenStreetMap (OSM). Both are cleaned and joined into one property table that feeds all three engines.

2.1. Data needed

To estimate a fair asking price, flag listings that stay online for more than 180 days and compare districts, the project needs three things: what the home is, where it is, and how long it has been listed.

Group	Variables	Type	Source	Why it is needed
Price (target)	Asking price (VND)	Number	mogi.vn	What the Fair Value Engine predicts
Home	Property type, area, bedrooms, bathrooms, storeys, frontage, plot depth, access road width, legal status	Number, category	mogi.vn	Main drivers of price
Text flags	Car access, corner lot, business use, lake view, near school, owner selling, urgent sale, furnished	0/1	mogi.vn description	Facts not in the listing form

Seller	Broker profile or private account, number of photos, description length	Category, number	mogi.vn	Inputs to the Stale Listing Radar
Location	City, district, ward, latitude, longitude	Text, number	mogi.vn	Places the home; used to join map data
Nearby places	Distance to the nearest metro station, bus stop, school, university, hospital, park, market, water and city centre; counts of bus stops, schools and markets within 500 m and 1 km; one accessibility index	Number	OSM	Measures access to amenities
Listing age	Days since the listing was created; stale = more than 180 days	Number, 0/1	Derived from listing ID	Label for the Radar only, not an input

Exhibit 2.1.1. Variables used

The full data dictionary is in result/tables/data_dictionary.csv.

2.2. Data sources

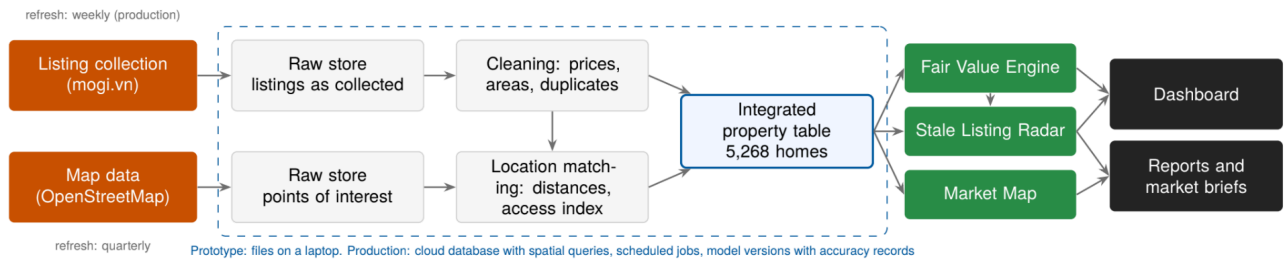
Source	Content	Size	Planned refresh
mogi.vn sale listings (mogi's database)	Price, home attributes, address, map point, description	For prototype: 6,000 collected; 5,268 clean (2,638 Hanoi, 2,630 HCMC; 51 districts)	Weekly
OpenStreetMap	Metro stations (20 Hanoi, 14 HCMC), bus stops, schools, universities, hospitals, parks, markets, water	34,321 points in 8 layers, plus metro lines	Quarterly
GADM 4.1	District boundaries	Vietnam, level 2	Fixed file

Exhibit 2.2. Data sources

2.3. Data architecture

Exhibit 2.3 illustrates the data flow. Listings and map data are collected, cleaned and matched by location, then combined into a single property table of 5,268 listings. This table supports the Fair Value Engine, Stale Listing Radar and Market Map, which provide results for the dashboard and market reports.

The two sources share no ID, so they are joined by coordinates. The prototype stores files on a laptop; a production version would use a cloud database with spatial queries and scheduled jobs.



Source: team design.

Exhibit 2.3. Data architecture

2.4. Integration challenges

Challenge	Response
Prices, dates and areas written as text (e.g. "4 tỷ 760 triệu")	Parsers convert them to numbers. Parsed prices match the site's own numeric price on 100% of clean listings.
No shared ID between listings and map data	Join by coordinates; compute distances and counts within 500 m and 1 km
Same home posted by several brokers	Remove near-duplicates and keep the earliest (219 removed)
Missing values (storeys 54%, access road width 87%, plot depth 31%, frontage 29%)	Fill with the typical value for the property type and add a "missing" flag
Imprecise location	Treat distances as accurate to the ward; keep district names as listed
Posting dates reset when sellers re-post (59% re-posted more than 30 days after their photos)	Measure listing age from the listing ID, calibrated to the first photo date

Exhibit 2.4. Challenges and responses

Limits of the data. The data is enough for a prototype, not for lending. Prices are asking prices, not transaction prices, and all listings come from one portal on one day. Phase 2 adds bank appraisal prices and a second portal. Weekly collection would also show when listings disappear, giving a true time-to-sale label.

2.5. Data cleaning

Cleaning kept 5,268 of 6,000 listings (88%); Exhibit 2.5 shows each step.

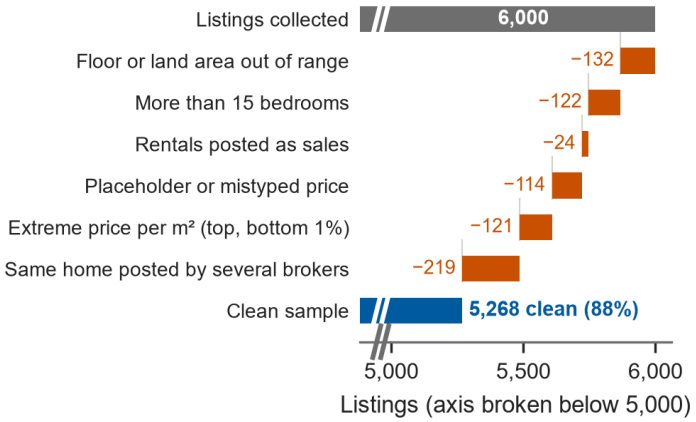


Exhibit 2.5. Cleaning waterfall

3. Method

3.1. Analytical Design

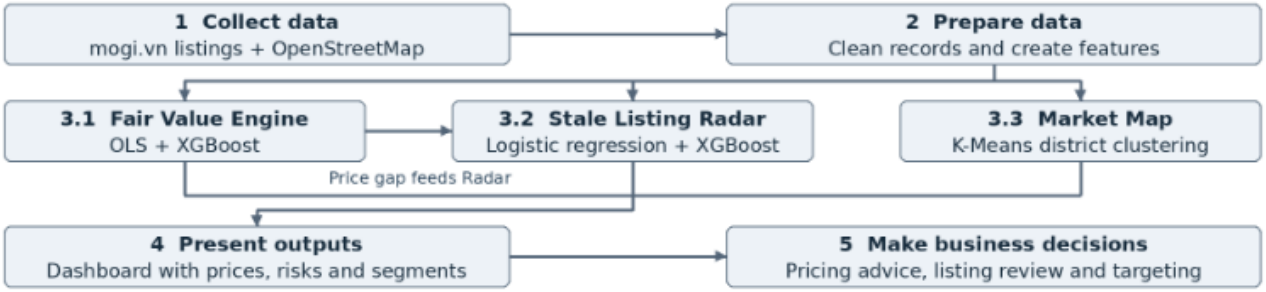


Exhibit 3.1. Analytical Design

Steps 3.1–3.3 are analytical branches; the valuation gap also feeds the Radar.

Use held-out evaluation and out-of-fold price gaps, and exclude age-related Radar inputs to prevent leakage.

3.2. Rationale for Method Selection

Fair Value Engine. Property valuation is framed as a regression task because the required output is a continuous price estimate. Ordinary Least Squares (OLS) hedonic regression is selected as an interpretable baseline that links asking prices to property and location attributes, while XGBoost is incorporated to capture nonlinear relationships and interactions that OLS may not fully represent. Unlike simple district-average pricing, both methods account for granular differences among individual properties. Furthermore, SHAP values are utilized to support model interpretability, and prediction intervals are provided to communicate estimation uncertainty effectively.

Stale Listing Radar. Classification is chosen for the Stale Listing Radar because the available labels distinguish listings above and below a specific age threshold in the absence of confirmed sale dates. A simple price-gap rule serves as a baseline business benchmark, complemented by logistic regression as an interpretable statistical baseline. XGBoost is also included to capture complex relationships among pricing, property, seller, and district characteristics. Ultimately, the ranking outputs enable managers to prioritize reviews efficiently in alignment with team capacity.

Market Map. Clustering is selected for the Market Map since the districts lack predefined segment labels. K-Means is employed to group districts based on price, listing-age, accessibility, housing-mix, valuation-gap, and volume profiles, delivering interpretable segments for targeted marketing and resource allocation, with features standardized due to varying measurement scales. Additionally, Ward hierarchical clustering is applied as an alternative grouping method to verify consistency across the results.

3.3. Output and Business Decision

Component and Output	Decision Owner	Business Decision
Fair Value Engine Price reference, prediction interval and comparables	- Brokers - Sales managers.	Advise on pricing and negotiation; review property details before recommending repricing.
Stale Listing Radar Staleness ranking and review list.	- Brokerage Managers	Prioritise contract and verification according to team capacity; a high

		score alone does not justify a price cut
Market Map Distinct groups and profiles	- Marketing Managers/ Executives	Adapt campaign and staffing; investigate target areas before committing investment

3.4. Evaluation Plan

Engine	Metric	Meaning	Threshold
Fair Value	RMSE log, MdAPE	Overall error and median % error	RMSE < 0.20
	PPE10, PPE20	Share of estimates within ±10% / ±20% of price	PPE20 ≥ 60% for the broker pilot; banks also need over-valuation above 20% in at most 10% of cases
	COD, PRD	Uniformity and price-related bias	IAAO 2013: COD 5 to 15 (older, heterogeneous residential), PRD 0.98 to 1.03.
	90% range coverage	Share of actual prices inside the range	90%
Stale Radar	AUC, PR-AUC	Ranking quality	AUC > 0.72
	Precision and capture in the top 10%	Accuracy when only the riskiest group is reviewed	Above the base rate
	Brier score, calibration	Whether probabilities can be trusted	Below logistic regression
	Expected cost by threshold	Threshold set by the FP:FN cost ratio	1:1, 1:3, 1:5
Market Map	Silhouette	Separation between segments	> 0.45
	Adjusted Rand Index vs Ward	Stability across algorithms	0.65 or more is moderate

3.5. Prototype results

The prototype meets two of the four project targets (OLS R^2 and Radar AUC). Figures are measured on unseen data (test set or out-of-fold), except the OLS R^2 and the silhouette, which use the full sample.

Metric	Target	Result	Met?
OLS R^2	> 0.75	0.850	Yes
XGBoost RMSE, log price	< 0.20	0.354	No

PPE20 (±20% of asking price)	≥ 60% (pilot gate)	53.9% overall; PPE10 29.8%; median error 18.1%	Apartments and alley houses only
Radar AUC	> 0.72	0.81 (95%: 0.78 to 0.83)	Yes
Market Map silhouette	> 0.45	0.21 (k = 4)	No

Exhibit 3.5. Prototype results against project targets

- Accuracy differs by type. PPE20: alley houses 64%, apartments 60%, villas 47%, land 43%, street-front houses 42%. Map data alone recovers 60% of the gain from knowing the district.
- The Radar beats its baselines. AUC 0.81 against 0.74 (logistic regression) and 0.45 (price-gap rule); the top 10% of scores are 99% stale against 65% on average. The price gap ranks only 12 of 26 drivers, so stale listings are not mainly overpriced ones.
- Four segments over 34 districts (Prime core, Apartment mid-market, Townhouse mid-market, Peripheral growth). With a low silhouette and ARI 0.47 against Ward, they are targeting groups, not natural borders.
- Limits for lenders. Expensive homes tend to be under-valued (PRD 1.12, outside the IAAO range of 0.98 to 1.03). ROI depends mainly on broker adoption, from −VND 2.3 bn to +40.9 bn a year (Exhibit 5.2).

4. Tools

4.1. Selection Criteria

The tools were selected based on six criteria: scalability, collaboration and reproducibility, usability for non-technical managers, cost-effectiveness, explainability, and data privacy. These criteria ensure the system can handle growing data, support teamwork, provide accessible and understandable insights, remain affordable, and protect user information.

4.2. Tools by Process Step

This exhibit lists the tool behind each layer and why it fits:

Layer	Tools	Outputs	Why It Fits
1. Data Collection	Python (requests, BeautifulSoup); OpenStreetMap via Overpass API; GADM	Raw listings, amenity points, administrative boundaries	Scale & Cost: resumable scraping reached 7,792 listings/hour in the pilot. OpenStreetMap data reduces dependence on paid map APIs; quarterly refreshes are planned.
2. Storage	Prototype: CSV, JSON, Parquet. Planned: cloud database with spatial queries	Integrated dataset of 5,268 homes	Scalability: local files avoid unnecessary infrastructure costs; cloud storage would support shared access, spatial queries and permissions.
3. Processing & Modeling	pandas, GeoPandas, statsmodels, scikit-learn, XGBoost, SHAP	Fair-value estimates with 90% prediction intervals; stale-listing risk; district segments	Explainability: regression provides a transparent baseline, XGBoost improves prediction, and SHAP explains individual estimates.

4. Presentation	Streamlit; standalone HTML/JavaScript; matplotlib, Folium	Valuation and market dashboards, district summaries	Usability: managers access predictions without coding. HTML produces near-identical results to Python on 500 test listings, without a separate Python deployment.
5. Operations & Governance (cross-cutting)	Prototype: run_all.py, pytest. Planned: scheduling, monitoring, model approval	Reproducible pipeline and tests; planned model oversight	Collaboration & Reproducibility: automated tests support reproducibility, and planned approvals govern production model changes.

Exhibit 4.1. Analytics architecture: selected tools, outputs, and business fit

5. Communication & Strategy

5.1. Value and ROI

The base case estimates VND 11.9 billion in annual partner benefits against VND 2.2 billion in operating costs, generating VND 9.7 billion in net value (Exhibit 5.1). Broker time savings account for 91% of these benefits, while reduced bank appraisal costs contribute the rest. Buyer savings are excluded from the return calculation. These figures are scenario estimates rather than measured financial returns, and first-year benefits will be lower due to gradual adoption. Adoption is the key driver (Exhibit 5.2): the conservative scenario loses VND 2.3 billion annually, while break-even requires about 30 active brokers under base assumptions or about 480 under conservative assumptions (Exhibit A.3, Appendix A). Therefore, broker adoption and usage must be closely monitored.

The strategic value goes beyond money. It would be the first public valuation of all property types in the two cities. The weekly listing panel becomes a price index and time-on-market record. A later entrant cannot rebuild that history, but it can copy the method, so the lead lasts only if broker usage and transaction calibration build on it (see Strategic advantage).

Input	Cons.	Base	Opt.
Brokers using the tool	100	300	500
Valuations per broker a month	10.0	20.0	25.0
Hours saved per valuation	0.50	1.00	1.25
Broker cost (VND m/hour)	0.10	0.15	0.20
Mortgage files pre-screened a year	1,000	3,000	6,000
Estimates within ±10% (PPE10)*	30%	30%	30%
PPE10 files where the bank waives appraisal	30%	50%	70%
Cost of an appraisal (VND m)	1.50	2.50	4.00
Run cost, 3-person team (VND m/year)	3,000	2,200	1,600

Buyer deals guided a year	300	1,000	2,500
Listings above the 90% range*	4.5%	4.5%	4.5%
Median excess above the range*	12%	12%	12%
Share of excess negotiated away	10%	20%	30%
Median asking price (VND bn)*	9.60	9.60	9.60

Annual value, VND bn	Cons.	Base	Opt.
Broker time saved	0.6	10.8	37.5
Bank appraisal cost avoided	0.1	1.1	5.0
Run cost	3.0	2.2	1.6
Net partner value	−2.3	9.7	40.9
Estimated net benefit-to-cost ratio (net value / run cost)	−0.8x	4.4x	25.6x
Buyer overpayment avoided (not in ROI)	1.6	10.4	39.0

Exhibit 5.1. Base case: partner value of VND 11.9 bn a year leaves net value of 9.7 bn, 4.4 times the run cost; the conservative case does not cover its cost [1]

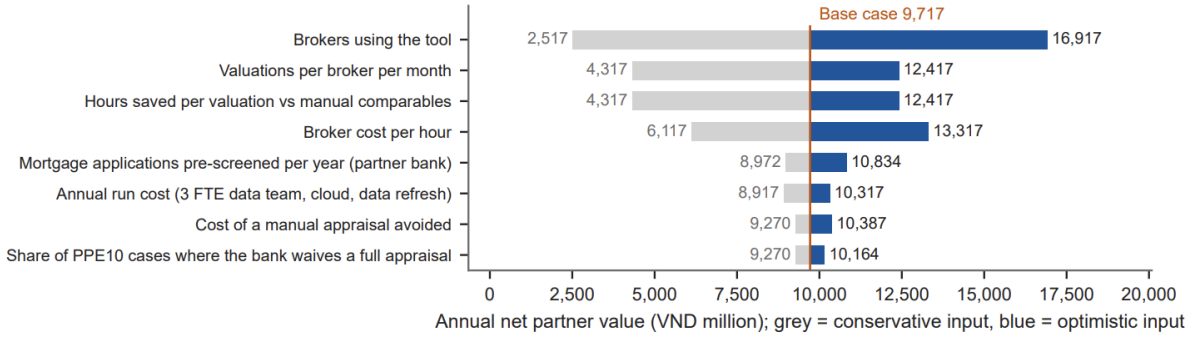


Exhibit 5.2. Broker adoption drives the business case more than any accuracy metric [2]

5.2. Strategic advantage

The platform's advantage is its position: the only public listing-level valuation, with a published method and a stated error. There is no other tool in the two cities that combines all three.

[1] Source: * measured in this analysis;other inputs are team assumptions, with market references in Appendix A; Net benefit-to-cost ratio = (partner value − run cost) / run cost.
[2] Source: team assumptions in Exhibit 5.1; each bar varies one input between its conservative and optimistic value, holding the others at base.

Advantage	Why it matters
Fills the empty public, all-type position	OneHousing is public but covers listed apartment projects only. Citics covers all types but serves banks only. Rever and Propzy built brokerages, not valuation tools, and Propzy closed in 2022.
Transparent and auditable	Method and error are published by property type. Every estimate shows a 90% range, which covers 90.5% of test homes, and five comparables. Broker judgement has no error band.
Works where comparables are thin	Map distances recover about 60% of the accuracy gain from knowing the district, so thinly listed districts can still be valued.
Three decisions from one table	One property table feeds price (Fair Value Engine), priority (Stale Listing Radar) and targeting (Market Map). Each department is involved with at least two engines.
A data asset that compounds	The weekly listing panel becomes a price index and a time-on-market record. Delisting events give an observed time-to-delisting measure, and transaction prices improve calibration in Phase 2. A verified time-to-sale label needs transaction or broker confirmation.
Lean cost base	Open-source tools and a 3-person team run the platform at VND 2.2 bn a year. Propzy, a capital-heavy model, closed in 2022 after failing to raise capital.
A reference for human decisions	Zillow's off-market error is several times its on-market error, and it closed its home-buying arm in 2021. The platform gives brokers a range, not an automatic bid, and keeps weak types on desk review.

Exhibit 5.3. Seven advantages, each tied to evidence in this report

These advantages hold today for apartments and alley houses, which reach 60% and 64% within ±20%. For street-front houses, villas and land, the advantage is coverage and transparency, not yet accuracy.

The data-asset advantage also starts only when weekly collection runs in production. The prototype runs on one machine with one listing source; cloud storage, scheduling, monitoring and access control are not yet built.

5.3. Implementation roadmap

Twelve months, three phases, each closed by a measured gate (Exhibit 5.4).

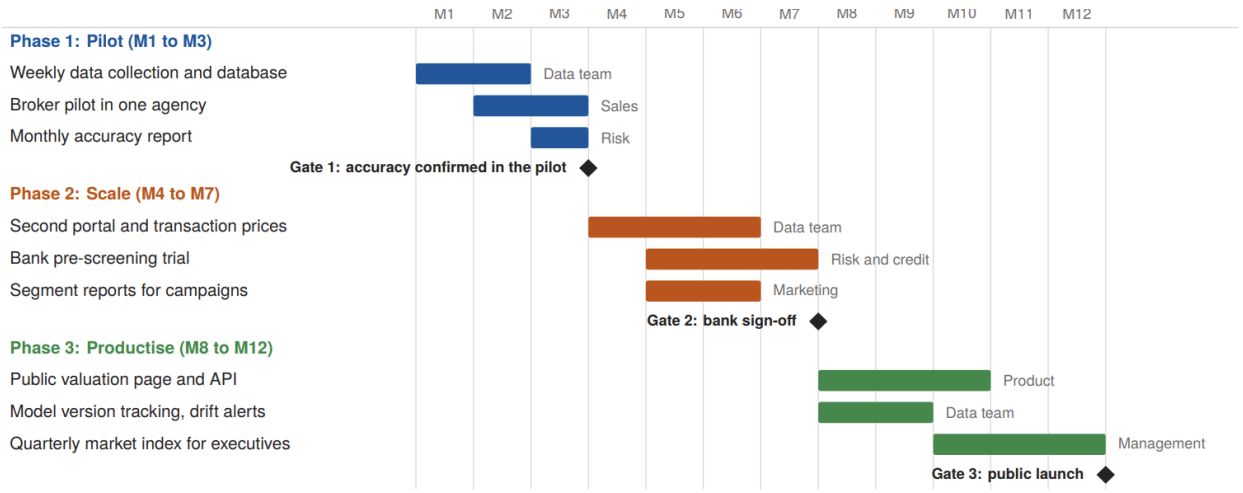


Exhibit 5.4: Twelve months in three phases, with a go or no-go gate at the end of each phase

Phase	Months	Activities (owner)	Output
1. Pilot	M1 to M3	Weekly data collection and database (Data team); broker pilot in one agency (Sales); monthly accuracy report (Risk)	First observed time-to-delisting measures; accuracy on pilot deals by type; weekly usage by pilot brokers
2. Scale	M4 to M7	Second portal and transaction prices (Data team); bank pre-screening trial (Risk and credit); segment reports for campaigns (Marketing)	Accuracy re-measured on appraisals or notarised prices; waiver rule agreed with the partner bank; 300 active brokers
3. Productise	M8 to M12	Public valuation page and API (Product); model version tracking and drift alerts (Data team); quarterly market index (Management)	Public page with accuracy by type; first quarterly index

Exhibit 5.5. Phases, owners and outputs

The roadmap has three measurable gates. By month 3, pilot accuracy for apartments and alley houses must remain at least 60% within ±20%, with regular broker usage. By month 7, the partner bank must validate estimates against its appraisals and agree on a waiver rule, while active brokers reach 300. By month 12, accuracy must remain stable for three consecutive months, and legal reviews must be completed before public launch.

5.4. Risks and Mitigation Plan

Exhibit 5.6 lists the six risks and their mitigation.

Risk	L / I	Mitigation
Listing site changes its pages or blocks collection	M / H	Tested page readers; adaptive pacing; second portal in Phase 2; negotiate a data feed

Asking prices differ from transaction prices	H / H	Calibrate on bank appraisals or notarised prices in Phase 2; present results as asking-price references until then
Stale label is noisy (refreshed dates, unseen sales)	H / M	Creation-date labels now; weekly collection gives an observed time-to-delisting, and transaction or broker confirmation is needed for a verified time-to-sale label
Accuracy drifts as prices move	M / M	Monthly accuracy report by type; re-train when PPE20 falls 5 points below launch
Legal and reputational risk of publishing valuations	L / H	Ranges, not single numbers; disclaimers and stated limits of use on every estimate; no personal data; published method and accuracy; review of listing-site terms and data-use rights before Phase 2; legal approval before the public API; a channel for users to challenge an estimate; model version and accuracy history documented
Brokers adopt the tool too slowly, or use it too little: the pilot covers one agency but the case needs 300 active brokers	M / H	Report active brokers and valuations per broker next to accuracy every month; build the estimate, range and comparables into the weekly listing routine; Gate 2 requires the 300 target, and if it is missed, add agencies before starting the bank trial; track against break-even (about 30 brokers at base inputs, about 480 at conservative inputs)

Exhibit 5.6. Six risks, each with a mitigation; the asking-price gap and the stale label matter most for the current results [3]

5.5. Recommendation

The Fair Value Engine should launch first for apartments and alley houses, with accuracy rates of 60% and 64% within ±20%. Street-front houses, villas and land should remain under manual review until accuracy improves. The Stale Listing Radar should prioritise listings for review rather than identify overpricing, while weekly data collection will help measure time to delisting. Before lending use, estimates must be validated against appraisal or transaction prices. Broker adoption remains the main business KPI, with a target of 300 active brokers by Phase 2 and monthly tracking of usage and accuracy.

5.6. RACI

The data team operates all three engines. The exhibit below assigns roles using the RACI matrix across departments and the three engines, ensuring that every model change is tied to a specific owner within business operations. Where: R = responsible (performs the work), A = accountable (owns the decision; one per column), C = consulted, I = informed

[3] Source: team analysis. L / I = likelihood / impact: H high, M medium, L low

Department	Fair value	Stale radar	Market Map
Sales and brokerage	R	A, R	I
Marketing and research	C	I	A, R
Risk and credit	A, R	C	I
Executive management	I	I	A
Data team	R	R	R

Exhibit 5.7. Each engine has one accountable department; the data team operates all three engines

Product builds and runs the public valuation page and API in Phase 3, working with the data team.

The production version will include model approval based on accuracy, monthly monitoring of performance and broker usage, and role-based access control without storing personal data. These features are not yet implemented. The current prototype runs on a single machine using one listing source, without operational-scale cloud storage, scheduling, monitoring or access control.

III. PROTOTYPE

The dashboard has two versions serving two distinct needs. The Streamlit version includes a Valuation page, where brokers read the results for an individual house, and a Market Overview page (executive dashboard matrix, district group profiles, and price tables by district). The standalone HTML version replicates the trained models using JavaScript, allowing it to be opened without installation or an internet connection; on over 500 random listings, its results strictly match those of the Python version.

Please click here to watch a video illustrating how the property valuation dashboard would work.

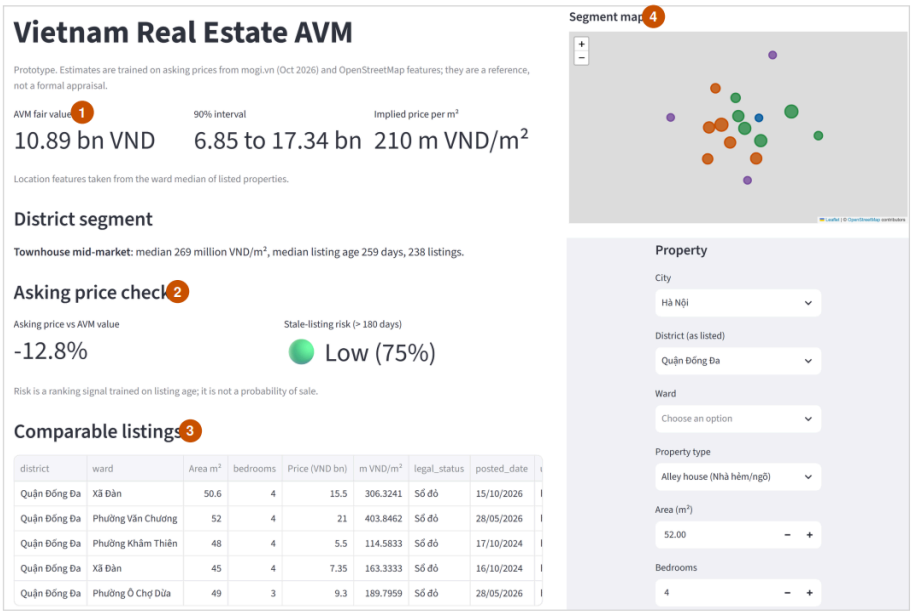


Exhibit III.1. A broker enters a home and an asking price and reads the fair value, its range, the price gap, the stale-risk light, comparables and the district segment

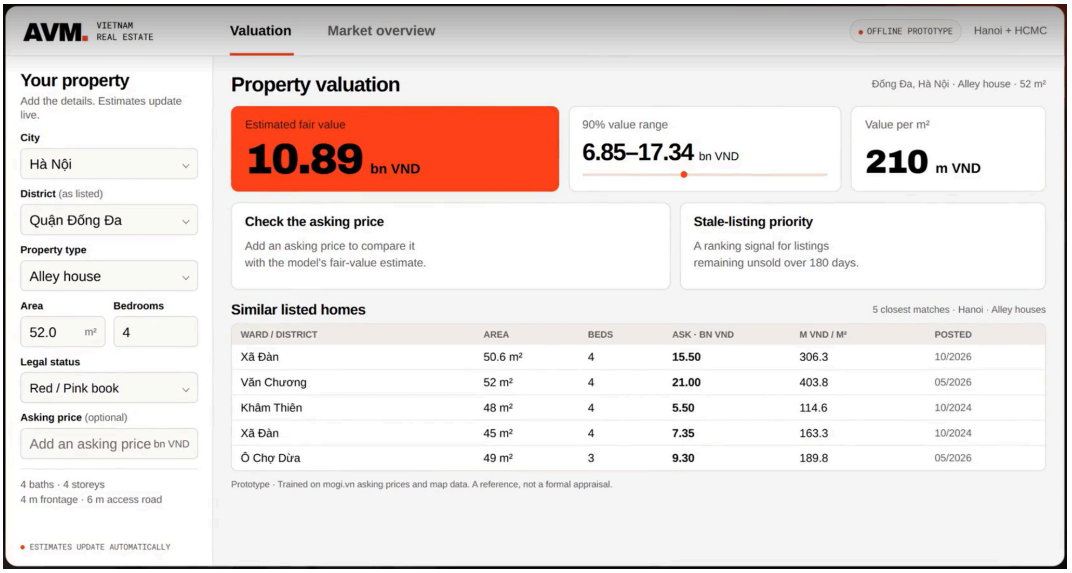


Exhibit III.2. The interface of the mock HTML Dashboard for property valuation

IV. CONCLUSION

This project demonstrates the potential of combining asking-price estimation, stale-listing detection and district comparisons to support property decisions in Hanoi and Ho Chi Minh City. While the prototype shows promising results for apartments and alley houses, limitations in data quality and model accuracy remain. Future development should prioritise transaction-based validation, model improvements and broker adoption before wider deployment. The platform is intended to support, not replace, professional judgement.

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APPENDICES

Appendix A. Assumptions and References for Exhibit 5.1

Of the fourteen inputs in Exhibit 5.1, four are measured on the 5,268 mogi.vn listings (1,054 test homes), six rest on market references (two on United States benchmarks only) and four are team assumptions that only the pilot can test. The references show whether a value is plausible; they do not prove it. ROI means the estimated net benefit-to-cost ratio of Section 5.1.

Input (conservative / base / optimistic)	Basis	Reference data and comment
Brokers using the tool: 100 / 300 / 500	Market reference	VnEconomy (March 2026): about 300,000 people in brokerage, over 40,000 certified; VARS via PLO: about 100,000 formal brokers. 300 brokers are 0.1% to 0.75% of these groups. National figures; no evidence yet that this many will adopt the tool.
Broker cost (VND m/hour): 0.10 / 0.15 / 0.20	Market reference	Báo Đầu tư (2022): about VND 15.4 m a month, over 30 m at large agencies; CafeF (Q2 2024): VND 13 m to 37 m. At 160 to 176 hours this is VND 0.09 to 0.19 m an hour, so the values lie in range. Income is mostly commission, so this is an opportunity cost.
Cost of an appraisal (VND m): 1.50 / 2.50 / 4.00	Market reference	HSBC Việt Nam fee schedule (2020): VND 2.1 to 4.5 m; independent valuers: VND 7 to 8 m for homes of VND 5 to 10 bn (ACC Group, LOTUS VFI). The base follows the bank schedule and is prudent; at VND 7.5 m the bank saving rises to about VND 3.4 bn.
Appraisal waived on PPE10 files: 30% / 50% / 70%	US benchmark only	June 2025: 11.5% of Fannie Mae and 17.1% of Freddie Mac loans waived, using AVMs on transaction data. The base implies 15% of files, close to Freddie Mac, on asking prices, so it appears optimistic. At 30% the net value is VND 9.3 bn and the conclusion holds.
Hours saved per valuation: 0.50 / 1.00 / 1.25	US benchmark only	No Vietnamese source. A 2022 survey (homegenius, Residential Real Estate Council): most US brokers spend over 30 minutes on a market analysis even with an MLS; Vietnam has none. Base break-even needs only about 6 minutes. The pilot measures the actual hours (Gate 1).
Run cost, three-person team (VND m a year): 3,000 / 2,200 / 1,600	Market reference (salaries)	CareerLink (March 2026): analyst VND 15 to 25 m, engineer 25 to 45 m, scientist 30 to 50 m a month; ITviec 2025-2026: engineer with 3 to 4 years about 56.9 m. Salaries for three people cost VND 0.84 to 2.05 bn a year; the base adds cloud services and data refresh.
Valuations per broker a month: 10 / 20 / 25; mortgage files: 1,000 / 3,000 / 6,000; buyer deals: 300 / 1,000 / 2,500; excess negotiated away: 10% / 20% / 30%	Team assumption	No market data found. Measured only in the pilot (Phase 1) and the bank trial (Phase 2); valuations per month and hours saved first, because break-even depends on them.

PPE10 30%; listings above the 90% range 4.5%; median excess 12%; median asking price VND 9.60 bn	Measured	Same in all cases. PPE10 is 29.8%, shown as 30%.

Exhibit A.1. Fourteen inputs of Exhibit 5.1: four measured, six with market references, four team assumptions [4]

Using the reference values does not change the conclusion (Exhibit A.2), and break-even depends on the bank saving (Exhibit A.3). All other inputs stay at base.

Case (base inputs)	Bank saving (VND bn)	Net partner value (VND bn)	ROI
Exhibit 5.1 base: waiver 50%, appraisal VND 2.5 m	1.1	9.7	4.4x
Waiver 30% of PPE10 files	0.7	9.3	4.2x
Appraisal VND 7.5 m	3.4	12.0	5.4x
Both changes	2.0	10.6	4.8x

Exhibit A.2. Using reference values, base net partner value ranges from VND 9.3 bn to 12.0 bn and ROI from 4.2 to 5.4 times [5]

Break-even threshold	Base inputs	Conservative inputs
Active brokers, bank saving counted	about 30	about 480
Active brokers, bank saving excluded	about 61	about 500
Hours saved per valuation, bank saving counted	0.10 hours (6 minutes)	2.4 hours
Hours saved per valuation, bank saving excluded	0.20 hours (12 minutes)	2.5 hours

Exhibit A.3. Break-even requires about 30 active brokers with the bank saving and about 61 without it, at base inputs [6]

Appendix B. Basis of Evaluation Metrics

Engine	Metric	Threshold	Basis
Fair Value	RMSE log, MdAPE	RMSE < 0.20	Internal target, no industry standard. If log errors are normal, RMSE 0.20 means about 69% of estimates within

[4] Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); sources named in the table, full citations in the References; team analysis.
[5] Source: team calculation from the inputs of Exhibit 5.1; ROI = net partner value / run cost.
[6] Source: team calculation from the inputs of Exhibit 5.1. The bank saving depends on the Phase 2 bank trial succeeding.

			±20% and 38% within ±10% (Steurer et al. 2021 on metric choice).
Fair Value	PPE10, PPE20	PPE20 ≥ 60% for the broker pilot; for banks, over-valuation above 20% in at most 10% of cases	60% is the test-set level for apartments (60%) and alley houses (64%), so the gate is a no-degradation condition. At a 70% loan-to-value ratio, a 20% over-valuation lifts the loan to 84% of true value; the cap keeps that to one file in ten. The 2024 US AVM rule sets no numeric threshold.
Fair Value	COD, PRD	IAAO 2013: COD 5 to 15, PRD 0.98 to 1.03 (now 27.0 and 1.12)	Reference only: IAAO is written for assessed value / sale price, here applied to AVM / asking price.
Fair Value	90% range coverage	90% (now 90.5%)	Nominal level chosen in advance; split conformal guarantees it without distributional assumptions (Lei et al. 2018).
Stale Radar	AUC, PR-AUC	AUC > 0.72, judged on the lower bound (now 0.78)	Hosmer, Lemeshow & Sturdivant (2013): 0.70 to 0.80 acceptable, 0.80 to 0.90 excellent; 0.72 is the floor plus a small margin.
Stale Radar	Precision and capture in the top 10%	Above the base rate (lift > 1)	A random review has precision equal to the base rate (65%), so beating it beats random selection.
Stale Radar	Brier score; expected cost by threshold	Brier below logistic regression (0.176 vs 0.189); cost ratios 1:1, 1:3, 1:5	Brier has no absolute threshold, so it is compared with the benchmark. The cost-optimal threshold is c_FP / (c_FP + c_FN): 0.50, 0.25, 0.17 (Elkan 2001); ratios are illustrative, and misses usually cost more than false alerts.
Market Map	Silhouette	> 0.45; 0.51 or more is better	Common scale: 0.26 to 0.50 weak, 0.51 to 0.70 reasonable. The 0.45 target sits in the weak band.
Market Map	Adjusted Rand Index vs Ward	0.65 or more is moderate (now 0.47: poor)	Common scale: 0.90 excellent, 0.80 good, 0.65 moderate, below 0.65 poor.

Exhibit B.1. Basis of evaluation metrics: thresholds and where they come from [7]

Appendix C. KPIs and Metrics

Baselines come from the prototype; scenario inputs are assumptions (Exhibit 5.1); proposed targets require pilot validation and approval by the business owner.

KPI	Baseline	Target or trigger	Owner, frequency

[7] Links to the sources named above go in the References.

PPE20 by type (within ±20%)	Apartments 60%, alley houses 64%	At least 60% on pilot deals (Gate 1); re-train 5 points below launch	Data team, monthly
PPE10 and 90% range coverage	29.8%; 90.5%	Re-measure against appraisals before lending use (Gate 2); range near 90%	Data team, quarterly
Radar AUC and top-10% lift	0.81; 1.5 times	Above 0.72; not below 1.5 times	Data team, monthly
Active brokers (one valuation a week)	0	Weekly use in pilot (Gate 1); 300 by end of Phase 2 (Gate 2)	Sales, monthly
Valuations per broker a month; hours saved per valuation	Not measured	20 and 1.0 (base); 10 and 0.5 (conservative); measured in pilot	Sales, monthly
Median days to delisting, coached vs uncoached	Proxy: median listing age 299 days	Coached shorter; number set after Phase 1 baseline	Sales, quarterly
Leads per listing, by segment	Not measured	Set after the first campaign	Marketing, per campaign
Pre-screens confirmed within ±10% by final appraisal	Not measured	Agreed with partner bank at Gate 2	Risk and credit, monthly
Change in median listing age by segment	Prime core 328, apartment belt 364, townhouse belt 254, periphery 278 days	Falling in apartment belt (proxy until time-to-delisting exists)	Management, quarterly
Net partner value and run cost a year	VND 9.7 bn; VND 2.2 bn (base scenario)	Net value above zero from Gate 2; run cost at most VND 2.2 bn	Management, quarterly
Weekly refresh completed; share of listings with time-to-delisting	One prototype run; 0%	Every week completes; proxy replaced within three months	Data team, weekly

Exhibit C.1. Eleven KPIs with baseline, target and owner

Three decide the gates: PPE20 by type (Gate 1), active brokers and the bank comparison (Gate 2), and the published accuracy history (Gate 3).